KINES/HEDU/PRT 7103: Design and Analysis 1

Spring 2020
Tuesday 4:35-7:05pm, HPER North 229
3 Credit Hours

Instructor: Keith Lohse, PhD, PStat
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Office Hours/Location: HPER N 258, drop In: Thursdays from 2-4pm
(Other times by appointment.)

Course Description:

Students will acquire working familiarity with the basic principles and theory of descriptive and inferential statistics. Topics will include measurement, elementary sampling, issues in data collection, data description, association, probability, and fundamentals of hypothesis testing and inference. Focus is on conceptual understanding rather than mathematical computation.

Pre- or co-requisites:
Prerequisites: KINES 7102 OR H EDU 7102 OR PRT 7102.

Required Materials:

- Judd, C. M., McClelland, G. H., & Ryan, C. S. (2017). Data Analysis: A Model Comparison Approach to Regression, ANOVA, and Beyond. (3rd Edition.) Routledge.
- In the computer lab, we will be using the open-source programming language R, and programming environment R Studio. Students are also encouraged to download this free software to their own computers.
- Video lectures and assignments will be posted on Canvas and available through the course web page: https://sites.google.com/site/lohsekr/design-and-analysis-i

Recommended Supplemental Materials:

- Field, A., Miles, J., & Field, Z. (2012). *Discovering Statistics using R*. Sage.
- "Discovering Statistics" is a valuable supplement to the "Data Analysis" textbook. While the Data Analysis textbook focuses on theory, the Discovering Statistics text is focused on application. Discovering statistics is also the textbook that students will use in KINES/H EDU/PRT 7104. If you would like to "try out" one or both books before purchasing, I have copies available in my office.
- Note that I am also happy to work with you on your own data using the Discovering Statistics textbook. Again, the focus of this book is on how to use R to do statistical analysis rather than the math underlying the statistical analysis itself. If you ever have data you want to work on, please feel free to bring it to office hours or schedule a meeting with me.

Expectations:

- **Of my students –** I expect my students to complete all their work outside of class in a timely manner (e.g., reading the book, watching video lectures). Completing these assignments before class allows students to get the most out of the “flipped-classroom” environment. In class, I expect students to be proactive in asking questions and respectful of each other. Working together in class is a large part of the course. By working together to complete assignments, students gain the benefits of asking questions, rapid feedback, and peer-to-peer teaching.

- **Of myself –** My students can expect me to be available, enthusiastic, and patient in presenting information, providing feedback, and working with them on their questions. Statistics is a challenging topic and students approach it from many different backgrounds. I will present explanations and examples that I think make sense to the broadest cross-section of students. However, students should expect that I can tailor my explanations to their background. In order to provide more personalized instruction, students can expect that I will be available to meet with them one-on-one. To that end, I maintain regular drop-in office hours, but I am also free to meet at other times by appointment.

Student Learning Outcomes

By the end of this course, you will be able to:

- Clearly and concisely describe different study designs.
- Choose appropriate analytical methods given their study design and research question.
- Describe foundational statistical concepts in the *general linear model* (e.g., measures of centrality, spread, and different probability distributions).
- Accurately explain the p-value, “statistical power”, and the relationship of “statistical significance” to both of these concepts.
- Describe the assumptions underlying the general linear model and perform diagnostic tests of these assumptions.

Teaching and Learning Methods:

As you will see, this course treats learning design and analysis like a game. "Statistics" is an unknown world that the student must explore. Hence, we pay homage to one of my favorite games by reminding students that, "**It's dangerous to go alone.**" If you are not familiar with the Legend of Zelda; in the beginning of the game the protagonist, Link, is given a sword to start his journey and told, “It’s dangerous to go alone! Take this.” and the old man gives him a sword.

[The Legend of Zelda, 1986]

Over the course of his adventures, Link not only learns how to use this sword more effectively, but learns to use a host of other weapons as well. Similarly, we start with learning some very basic statistics and how to implement them in R. As our skills grow, however, we will start to

learn to use other packages and programs to deal with different types of data and particularly difficult analyses.

To achieve these aims, we will use a “flipped-classroom” model in which students are expected to do the readings and watch pre-recorded lectures outside of class. In class, students will work alone or in small groups to complete assignments for that week, putting the knowledge they gained from those assignments into practice.

Assignments:

The course begins with all students having zero experience points (XP). Starting from zero, you can gain points in different ways throughout the course.

You can earn XP in three ways.

Online Quizzes (20% of Total Grade). First, you can complete weekly quizzes that will be given online through Canvas prior to the beginning of class.

Assignments (40% of Total Grade). You can earn XP from lab assignments that require you to implement the knowledge you have gained outside of class. These assignments are to be done in lab with support from the instructor. These skills will prepare you for the quests in the course and (hopefully) for the application of statistics in your own research. These assignments must be completed in class, so it’s important for you to attend class as often as possible (otherwise you will miss out on XP!).

Quests (40% of Total Grade). To earn XP from a quest, you will need to complete the practical module that goes with that quest with at least 80% accuracy. To start a quest (more information will be given in class), you can download the .pdf with all of the questions for that module from Canvas. You can complete the module at home in the allotted time frame using any of the materials from the course and then upload the completed quest through Canvas. All students are expected to work on the quests individually. You can certainly study other course materials together, but the quests should represent your independent work.

Grading Criteria:

Grades in the course will be based on the following numeric cut-offs. Please note that I will not round final grades in this course. Students have opportunities for partial credit on all assignments and all exam questions are subject to an item analysis (with massively missed or poorly written question dropped from the tabulation of your grade).

Minimum	Grade	Minimum	Grade	Minimum	Grade
92.5	A	79.5	B-	66.5	D+
89.5	A-	76.5	C+	62.5	D
86.5	B+	72.5	C	59.5	D-
82.5	B	69.5	C-	0	E

If students have questions (following my written feedback), I am always happy to meet with students and discuss their overall grade or their grade for individual items in more detail.

Course Schedule:

Note: Our schedule is subject to revision based on the needs of our class; any updates will be made available for students through Canvas.

Week	Date	Module	Topics, Readings, Assignments, Due Dates
1	01/07	01	Introduction to Data Analysis
2	01/14	02	Definitions and Parameter Estimates
3	01/21	03	Models of Error and Sampling Distributions
4	01/28	04	Statistical Inferences about Parameter Values
5	02/04	05	Estimating Models with a Single Continuous Predictor ***(Quest 01 Assigned; Covers Modules 1-5).***
6	02/11	06	Models with Multiple Continuous Predictors
7	02/18	07	Moderated and Nonlinear Regression ***(Quest 01 Due.)***
8	02/25		(*continued*)
9	03/03	08	Models with a Single Categorical Predictor
10	03/10		*****Spring Break*****
11	03/17	09	Models with Multiple Categorical Predictors
12	03/24		(*continued*)
13	03/31	10	Models with Non-Independent Errors (Repeated Measures)
14	04/07		(*continued*)
15	04/14	11	Outliers and Ill-Mannered Error (Regression Diagnostics) ***(Quest 02 Assigned; Covers Modules 6-11).***
16	04/21		(*continued*)
17	***04/28***		***(Quest 02 Due.)***

Course Policies:

Food & Drink: As this course is taught in a computer lab, no food or 'uncovered' drinks are allowed. Any drinks must be in a sealed container that will be resistant to spilling even if tipped over, dropped, etc. (e.g., thermos, travel mug).

Canvas: Due to the Federal Educational Rights and Privacy Act, all electronic correspondence must be sent through Canvas or your official University of Utah (U-Mail) account. In addition, all official communication for the course (e.g., a cancelled class) will be made through Canvas. Please check for updates regularly.

University Policies:

1. ***The Americans with Disabilities Act***. The University of Utah seeks to provide equal access to its programs, services, and activities for people with disabilities. If you will need accommodations in this class, reasonable prior notice needs to be given to the Center for Disability & Access (CDA; http://disability.utah.edu/; 162 Olpin Union Building; 801-581-5020). CDA will work with you and the instructor to make arrangements for accommodations. All written information in this course can be made available in an alternative format with prior notification to the CDA. *Accommodations can only be made once the student has gone through the application and assessment process and the staff have indicated that an accommodation is warranted.*

2. ***Addressing Sexual Misconduct***. Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a Civil Rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status or genetic information. If you or someone you know has been harassed or assaulted, you are encouraged to report it to the Title IX Coordinator in the Office of Equal Opportunity and Affirmative Action, 135 Park Building, 801-581-8365, or the Office of the Dean of Students, 270 Union Building, 801-581-7066. For support and confidential consultation, contact the Center for Student Wellness, 426 SSB, 801-581-7776. To report to the police, contact the Department of Public Safety, 801-585-2677(COPS).

3. ***The Code of Student Rights and Responsibilities***. The code, which specifies student rights as well as conduct involving cheating, plagiarism, collusion, fraud, theft, etc., is provided at http://regulations.utah.edu/academics/6-400.php.

4. ***Wellness Statement***. Personal concerns such as stress, anxiety, relationship difficulties, depression, cross-cultural differences, etc., can interfere with a student's ability to succeed and thrive at the University of Utah. For helpful resources contact the Center for Student Wellness, http://wellness.utah.edu/; 801-581-7776.

5. ***Drop/Withdrawal***. The last day to drop classes is January 17th; the last day to withdraw from this class is March 6th. Please check the academic calendar for more information pertaining to dropping and withdrawing from a course. Withdrawing from a course and other matters of registration are the student's responsibility.

6. ***Campus Safety.*** "The University of Utah values the safety of all campus community members. To report suspicious activity or to request a courtesy escort, call campus police at 801-585-COPS (801-585-2677). You will receive important emergency alerts and safety messages regarding campus safety via text message. For more information regarding safety and to view available training resources, including helpful videos, visit safeu.utah.edu."

University Resources:

1. ***Veterans Center***. If you are a student veteran, I want you to know that the U of Utah has a Veterans Support Center on campus. They are located in Room 418 in the Olpin Union Building. Hours: M-F 8-5pm. Please visit their website for more information about what support they offer, a list of ongoing events and links to outside resources: http://veteranscenter.utah.edu/. Please also let me know if you need any additional support in this class for any reason.

2. ***LGBT Resource Center***. If you are a member of the LGBTQ community, I want you to know that my classroom is a safe environment. Additionally, please know that the

U of Utah has an LGBT Resource Center on campus. They are located in Room 409 in the Olpin Union Building; their hours are M-F 8-5 pm. You can visit their website to find more information about the support they can offer, a list of events through the center and links to additional resources: http://lgbt.utah.edu/. Please also let me know if there is any additional support you need in this class.

3. ***Learners of English as an Additional/Second Language***. If you are an English language learner, please be aware of several resources on campus that will support you with your language development and writing. These resources include: the Writing Center (http://writingcenter.utah.edu/); the Writing Program (http://writing-program.utah.edu/); the English Language Institute (http://continue.utah.edu/eli/). Please let me know if there is any additional support you would like to discuss for this class.

4. ***The Center for Disability and Access.*** *The Center for Disability and Access provides accommodations and services to students at the University of Utah whose condition significantly impacts their ability to function in the academic setting. Specific accommodations that are available to students depend on the documentation of the disability, the essential elements of the course, and the functional limitations created by the student's disability. In order to qualify for accommodation, students must schedule an appointment with CDA. More information about accommodations and services as well as how to qualify for services can be found on their website: https://disability.utah.edu*